

THE ROLE OF DASHBOARDS IN BUSINESS DECISION MAKING AND PERFORMANCE MANAGEMENT

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ABSTRACT

It is rightly said, "A picture is worth thousand words". When data is transformed into graphs and visuals, the informational dashboards provide picture of business performance in understandable and useful format which provides business insights for the decision makers to take timely decisions to run the business optimally. Business Intelligence (BI) dashboards support data visualization, performance management and monitoring. BI features like Predictive Analytics helps in data based forecasting built on historic data and various patterns. However, designing and choosing a perfect dashboard with effective data visualization is still a challenge and many organizations are still unable to make fruitful data driven decisions due to various factors. Still, enough literature is not available on the extent of BI dashboards & their effectiveness. In this paper, we conducted a comprehensive literature review to identify the challenges faced by users and organizations while implementing BI and investigated the BI Capabilities that makes BI dashboards effective for decision making. The study also included in-depth interviews of 15 professionals who have been widely using dashboards for **decision making** in various business domains. Interviews with senior managers and executive were focused on business needs for various purposes like **performance management**, Predictive Analytics, forecasting and strategic planning and their **expectations** for a perceived success. Useful **capabilities** and features of BI to support and improve the decision making for effective actions has been studied which helps to identify issues and challenges in BI implementation. These findings can serve as a guide for developers and users in designing better BI dashboards that help decision makers in making faster and effective decisions.

Keywords: Business Intelligence, Dashboards, Performance Management, Predictive Analytics.

1. INTRODUCTION

Quantity isn't quality. Decision-makers need more than information – they need insight states (Allio, 2006). Today's complex business World is ever changing at amazingly high speed and the fierce global competition and growing customer demands have compelled business organizations to constantly explore all possible ways to optimize the resources, deliver products and services on TIME. To remain competitive and meet customer demands, organizations continuously need reliable and relevant information or business insights at right time to right decision makers to enable them to take Effective and timely Decisions to achieve organizational goals efficiently. Business Intelligence (BI) is a technology-driven process of gathering, storage and analyzing data that is presented as actionable information to help business managers in making informed business decisions (Negash and Gray, 1989). BI transforms complex data into actionable information and business insights.

The main purpose of a BI dashboard is to support decision making (Presthus and Bergum, 2015). Organizational dashboards are just like the dashboards in aircrafts and cars, which provide at-a-glance information to the decision makers & helps in performance management, Key Performance Indicators (KPI) monitoring and scorecards.

Performance dashboards are a commonly used management tool for performance management and progress monitoring which assist the decision maker to make better decisions faster. BI Dashboard is a visual display of important information organized on a single screen so the information can be monitored at a glance (Few, 2004). Dashboards facilitate visibility and helps to analyze complex data with various trend analysis and scenario analysis options. Forward-looking companies are increasingly adopting sophisticated methods of analyzing employee data to enhance their competitive

advantage (Davenport, Harris, and Shapiro, 2010). Dashboards provide a way of easily communicating data at all levels of a business, and can reveal interesting patterns and relationships, which can be effectively used for predictive analytics. The key capabilities of dashboards include data visualization with graphs and maps, drill-down features, Predictive Analytics, mobile BI, and self-service BI. BI tools allow sharing information to the right people at right time using visual dashboards, scorecards & reports. Managers need smart and timely intelligence to make effective business decision and with complex data, data visualization and dashboards help to transform huge data into reliable intelligence.

Technological capabilities such as data quality, user access and the integration of BI with other systems are necessary for BI success (İşik, Jones, and Sidorova, 2013). User quality, information content quality were the significant factors contributing to use of information from the BI Systems (Sparks and Hollis, 2014). When all the basic expectations are met, BI indeed can transform complex data into business insights consistent with Microsoft's wise marketing tagline "Turn data into opportunity".

Very few studies however examine impact of effective dashboards on organizational performance management through effective decision-making. This research fills this

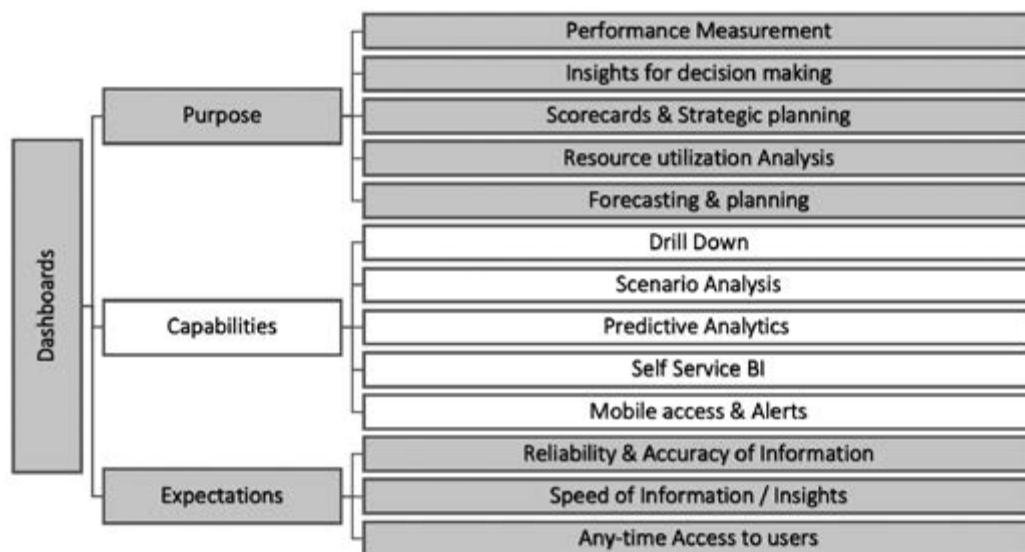
gap in knowledge by examining the practical implication of BI dashboards. This study is significant as it provides insights from perspectives from both BI developers as well as industry professionals who use BI for business insights. Our study addresses these research questions by 15 in-depth interviews conducted with BI developers / Analysts & Industry professionals. The industry professionals were senior managers & decision makers from Engineering and pharmaceutical industry, who validated some of the claims from IT professionals, highlighted purpose and expectations from the business and also highlighted some of the reasons for success or failure of BI. The research questions are to investigate if BI dashboards enable faster and better decisions and the relationship between usage of a BI & quality of data driven decisions made. It also explores the Capabilities of BI dashboards and how do they affect user productivity and finally try to investigate what characterizes an effective dashboard in organizations.

The next section briefly reviews the most important studies related to BI dashboards & its capabilities. This is followed by a methodological approach for answering the research questions. We then present our findings, synthesize them & finally conclude the paper with a discussion of our contributions and limitations of our study.

2. LITERATURE REVIEW

For the literature review, we searched papers based on keywords in research databases including Scopus, ProQuest, EBSCO, and Emerald & SpringerLink. Google Scholar was also extensively used for search using keywords: "Business Intelligence", "Dashboards", "Data Visualization", "Key Performance indicators", "KPI", "decision support systems" & "scorecard". The returned articles were filtered according to their relevance and were reviewed to present a summary of the most important findings relevant to dashboards.

Figure 1: Dashboards - Purpose, Capabilities and Expectations



Noonpakdee and Khunkornsiri, (2018) proposed framework comprising 4 main components - business operation, attribute list, visualization, and dashboard capabilities. Researcher consolidated various learnings to provide a quick overview in above Figure 1: Dashboards - Purpose, Capabilities and Expectations.

The dashboard capabilities are concerned with Dashboard layout, design, Presentation, Alerting, Analysis, Dashboard

interactivity, Delivery, and Architecture. The differences between various BI systems include the underlying architecture, how easily the organization's KPI's can be developed into metrics, their ability to "drill-down" into different levels, and the flexibility and presentation of the interface (Cox, 2011).

(Presthus and Bergum, 2015) conducted an explorative study to investigate the capabilities of BI dashboards and to understand how BI dashboards can be adopted by citizens in health care. They proposed a set of guidelines for BI dashboard adoption. Study by Presthus and Canales (2015) investigated as to what characterizes the state-of-the-art BI dashboard in organizations and how should it be designed to facilitate decision making. Based on six questions framed, they proposed framework comprising of: (1) number of KPIs, (2) layout, (3) drilling, (4) visual principles, (5) technical feature, and (6) contextual information.

The BI tools and dashboards should provide an accurate & timely information in a form easy to interpret and take action to improve competitive advantage of the organization (Bose, 2009). The successful implementation of a dashboard is complex and requires a step-by-step process to effectively plan, design, build and deploy a dashboard. (Orts, 2005). A dashboard is indeed an important tool for monitoring the organizational performance but to it is important that the data availability and quality must be secured (Chae, Olson, and Sheu, 2014) through right IT infrastructure. (İşik et al., 2013) suggest that technological capabilities such as data quality, user access and the integration of BI with other systems are necessary for BI success, regardless of the decision environment. (Yigitbasioglu and Velcu, 2012) states that the Performance dashboards might offer a remedy to the information overload problem by providing an all-inclusive package for performance management. Indeed, BI tools offers "Powerful insights - At your fingertips" as stated in marketing tagline of one of the BI tool vendors.

Dashboards have evolved from performance monitoring to more advanced analytical purposes, with new features such as scenario analysis and drill-down (Oana Velcu-Laitinen, 2013). Dashboards help in the performance management and could significantly reduce the information load problem by decreasing the number of tools that the management is required to use nowadays. Using the BI tools, the complex corporate level data can be presented professionally for various reporting needs like business Performance Management and Scorecards.

(Attaran and Attaran, 2018) states that predictive analytics has become an increasingly hot topic in analytics landscape as more companies realize that predictive analytics enables them to reduce risks, make intelligent decisions, and create differentiated customer experiences. Predictive analytics enable decision makers to predict future trends and scenarios based on historical data and various patterns. As a result, predictive analytics is gaining momentum in recent past especially in large organizations.

(Ronald William Riker Jr., 2019) states that some of the challenges in finding skill deficiencies relate to the educated design of business performance management, dashboard construction, visualization, and metric management. (Watson, 2015) highlighted the importance of organizational leaders to identify and understand the requirements of advanced data analytics. If the BI skill level is high, the Self Service BI can enable users to be more self-reliant and less dependent on power users (Lennert, Van Laere, and Söderström, 2019).

Table 1: Qualitative Research lists some literature related to qualitative research related to BI & analytics.

Title	Author	Findings
Understanding Business Analytics Success and Impact: A Qualitative Study	(Parks and Thambusamy, 2017)	Through a qualitative study, gained deep insights into the success factors and consequences of business analytics.
The implications of Big Data analytics on Business Intelligence: A qualitative study in China	(Ram, Zhang, and Koronios, 2016)	One of the main applications of Big Data analytics is for business intelligence to improve decision making capabilities, faster decision making.
The factors of failure in the implementation of dashboards as a tool to measure KPI: an exploratory qualitative inquiry	(Schwartz and SHARON, 2016)	The structure of a dashboard may be critical to the dashboard's success. A complete stakeholder support and user involvement is essential for BI success.

3. OBJECTIVES OF THE STUDY

In line with the suggestions of (Oana Velcu-Laitinen, 2013) for research focusing on the business value of dashboards in performance management, this study had focused interviews with the industry professionals probing to identify perceived business value of dashboards in performance management.

(Presthus and Bergum, 2015) suggested study to investigate optimum number of KPIs in a BI dashboard. During the study, the researcher observed various dashboards effectively used by some industry professionals for performance monitoring & KPI measurements. (Lea, Yu, and Min, 2018) in their study listed various KPIs used in Energy sector for performance management and many of these KPIs are common for most industries.

(Lennerholt et al., 2019) Suggested opportunity for further research to develop and formulate recommendations that could address each of the challenges identified. During this study, the researchers tried to identify the challenges as faced by the decision makers and then tried to get some ideas from the BI developers and IT professionals to find some solutions. (Schwartz and SHARON, 2016) after a qualitative study on “failure in the implementation of dashboards as a tool to measure KPI” suggested further research to study the factors of failure in the implementation of dashboards.

This study investigated user perception on both success and failure of BI dashboards and tried to identify the challenges faced by BI professionals as well as BI users. The purpose of this explorative study was to investigate the Capabilities of BI that can help the business managers to effectively use in various purposes like data driven decision making, performance management and the expectations of the business managers from BI system to add business value in their operations. These findings can serve as a guide for developers in designing a better dashboard. The results will help managers and business owners in putting place strategies to utilize BI dashboards better to understand their business and improve their decision making and profitability.

4. RESEARCH DESIGN & DATA COLLECTION

The aim of this paper was to identify the purpose of dashboards for decision making, performance management, KPI monitoring, communication, and analysis. It also identified the capabilities of BI that makes effective dashboards and the expectations for the BI success. This study was conducted by investigating several published papers related to BI.

Considering the exploratory nature of the research, we have adopted a qualitative approach to data collection and analysis. Based on an extensive literature review, the researchers have developed a semi-structured questionnaire for in-depth interviews of 15 IT and Industry professionals who have been widely using dashboards for decision making in various functional areas. Interviews with IT professionals and BI consultants were focused on technical design of the dashboards. Interviews with BI users and senior business managers were focused on understanding the key purposes of using BI dashboards, which BI capabilities are useful and what are the expectations for successful BI implementation. Each interview lasted between 60 to 90 minutes. The interview method allowed two-way communication that helped in seeking more detailed and in-depth information through open ended questions.

Focus of these interviews was to analyze current systems, integration with Enterprise resource planning (ERP) & other data and challenges to meet the end user requirements. Interviews also focused to study the impact of size of the organization, user level, and user knowledge and data quality. The research hypothesis were:

1. BI dashboards impact quality of data driven decision making
2. BI dashboards influence decision making speed.
3. Capabilities like drill-down, self-service BI, mobile access and interactive features influence success of BI.
4. BI dashboards impact performance management.
5. Predictive Analytics and trend analysis help in KPI monitoring & Performance management.

Motivated by some interesting insights provided by the BI developers, experienced BI users and managers, the researcher closely observed various dashboards to understand the capabilities, especially features like predictive analytics, drill-down, etc. and how the information is practically used in different organizations. Later, the researcher created some dashboards for various applications including sales analytics, vendor performance analysis, quality trends, on-time delivery analytics, shipping overviews, etc.

Table 2 : Data collection Instruments and Sources

Strategy	Data source and characteristics	Data	Remarks
Semi structured Interview	IT professionals & BI developers with more than two years' experience in dashboards	Field-notes	7 persons
Semi structured Interview	Industry professionals working in Engineering & Pharmaceutical Organizations & using dashboards	Field-notes	8 persons
Observation	The researcher critically observed six key dashboards as demonstrated by the BI developers & users		6 dashboards
Researcher Participation	The Researcher used the inputs from the BI developers and tried to validate the effectiveness personally on five different dashboards		5 dashboards

5. DISCUSSIONS & FINDINGS

In line with the Figure 1 : Dashboards - Purpose, Capabilities & Expectations, the researcher focused the interview questions in these 3 key areas to consolidate findings and draw the pattern to understand the role of BI dashboards in decision making and performance management of the organizations :

1. PURPOSE : We started exploring the key business areas where managers need BI dashboards. The study showed that most managers used BI tools to provide the business insights to help in decision making, KPI monitoring and timely communication to teams. Senior managers used BI for Performance management, KPI monitoring and strategic management professionals used them for business scorecards, resource optimization, strategic planning & demand forecasting. Below were some excerpts from the interviews:

A BI analyst claims, "The senior management professionals from our clients often ask us to provide dashboards in a balanced scorecard form. These kinds of dashboards are highly appreciated by the CEOs and CFOs and we have made various templates especially for these kinds of clients. The Scorecards generally have around 5 high level KPIs while the other performance dashboards could have 5-10 KPIs, depending on the domain and size of the organization."

"The predictive Analytics provide forecasts and insights about likely future outcomes which greatly help organizations working in challenging conditions and we help our clients by providing dashboards that help them to have a fair futuristic view / forecast based on the historic data pattern. Inventory forecast dashboards based on predictive analytics, for example, are highly appreciated by our clients handling complex supply chain operations," says a senior BI Analyst and adds -"We encourage our clients to use the mobile dashboard features and many of them use these features for their KPI measurements and communication across teams. Top management highly prefer scorecards for their strategic planning and performance management".

"We use balanced scorecard and dashboards have been very interesting in this area. It is much more efficient to have a single dashboard rather than managing multiple Excel sheets as we did in past" states an experience Engineering Entrepreneur. "We are tracking 5 KPIs to closely monitor our key goals".

"Predictive Analytics is a very useful feature, but its awareness is low. We strongly recommend our clients to take advantage of such useful capabilities. We have developed dashboards for forecasting their cash flows, warehouse utilization, expected sales, probable quality deviations, etc based on historic data and various patterns" states another BI Analyst.

"Predicting demand is helpful to prioritize and optimize the warehouse space," says a warehouse manager working in a pharmaceutical warehouse. "Our business is seasonal and our BI consultant is developing a dashboard to visualize our expected future goods movement and warehouse utilization and we very much look forward to it"

"BI dashboard monitoring has made our project management really very interesting," says a project manager of a large Engineering company. "Our projects are large and complex

and with the uncertainty, we need to monitor them closely even during our holidays. With mobile dashboard, I can take decisions much faster, even while enjoying a vacation," he adds with a spark.

"We used to use many Excel spreadsheets for our operational but KPIs and BI dashboards has simplified that so much, that we can spend lot of our time to work on the actions. The interactive feature of Power BI is just amazing and it gives us an opportunity to have a complete 360-degree view of our value chain," states a senior manager in a pharmaceutical company.

"I use BI to monitor the daily production levels at our remote site & that helped me a lot to save frequent travel," says a senior manager. "We used an automated packing system, integrated with high speed scanner and the BI dashboards really helped us in terms of performance management and KPI monitoring. With a focused monitoring, we managed to complete the complex project ON TIME".

2. CAPABILITIES : We next tried to explore as to which features and capabilities of BI are mostly helpful to achieve the business purpose. Most users stressed that the dashboard layout, design, interactivity and drill-down features have a great contribution to make BI dashboards successful in decision making, performance management, KPI monitoring and reporting. More experienced BI users liked the self-service BI and advanced features like scenario analysis and predictive analytics. Below are some excerpts from the interviews:

"The users like the interactivity options, filters and drill-down features. Different users have different preferences of colors and we need to spend sufficient time with the end user to design effective dashboard," says a BI developer. "Our clients in engineering industry appreciate Dashboards and interactive features. We simply aim to make BI dashboards easy to use".

"Data Visualization is one of the most important features and our users are highly excited when they look at some of the interesting visuals." - States a young BI developer.

"The Geographic map visual has been a great selling feature," says a BI professional. "CFOs and marketing managers simply love this feature, when we demonstrate the sales and profitability dashboards and these really help us closing the new business deals," she mentions with a smile.

"We need to have the exact batch details of every product and its expiry date for many decisions and visual dashboards with drill-down options makes our lives easy" says a warehouse manager working in a pharmaceutical warehouse. We can easily see a broad level overview with the warehouse utilization and then drill down to details based on a particular order or category.

"The dashboards with drill-down and search options gives us flexibility to view the trends and also look at current data. The export to Excel is very useful for us to do our planning. BI has greatly helped us to save our time and enabled us to deliver right products on right time to our esteemed customers and having timely quality information is critical in today's uncertain

business environment,” Says a planning manager responsible for the supply chain planning.

“The mobile alerts are very useful and gives me flexibility wherever I travel,” says a Sales manager from an Engineering Industry while an experienced Engineering Entrepreneur comments”

Manufacturing dashboards have helped us to save costs and I like the interactive visuals and drill-down features”.

“The self-service BI feature allows me to customize dashboard to my needs,” says one of the BI users working in a pharmaceutical warehouse operation.

3. EXPECTATIONS : Finally, we tried to investigate as to what the business managers expect from BI for a perceived BI success or failure. Most of the BI professionals stressed the importance of good Data Quality, integration with ERP, and user quality for a successful BI implementation. Some stressed that the dashboard layout, design & interactivity has a great contribution to make BI dashboards successful in performance analysis & reporting. Most of the decision makers felt that “anytime – anywhere data access”, Reliability and speed of the information and IT support were the key drivers for effective BI usage for data driven decisions and data-based performance management. Below are some excerpts from the interviews:

“Data quality & ERP integration is a key to provide an intelligent dashboard to senior managers” states a senior BI analyst. “Understanding business data is essential for making an effective dashboard and after spending sufficient time to discuss with the users and decision maker, I spend lot of time in relating the needs with the business data. With this approach, I could satisfy most of the clients. It is however difficult to please some users who simply resist the change and insist on traditional reports. Thankfully, BI also provides a

simple tabular data for such users”.

“Success of BI very much depends on the quality of BI users and data driven decision culture. Our clients who have included BI in their IT strategy have been more successful,” says a Senior IT professional. “Self Service BI feature is used by super-users who have been using Excel dashboards”

“Quality of data is a key for BI success & as such, the IT professionals should make data sources easy to access and use and ensure a seamless integration with ERP and CRM systems”. States an IT manager - he adds “It is important to define clear policy for data management and governance”.

“Our company introduced BI year back, but only very few persons were trained and there is lack of IT support for any queries. We tried making some dashboards, but many times, there has been lack of access which has been very disappointing,” claims a disappointed user from procurement support of a pharmaceutical company, who aspires to explore BI.

A warehouse user from pharmaceutical company stated, “We used to spend many hours to make report of our inventory and BI does not provide the correct values”. A further discussion with the IT indicated some issues with the ERP integration.

“Few users have some resistance to new technologies like BI , but simple discussions and demonstration of interactive dashboards helps to convince them for the use of BI” as per a senior IT professional working in a large Engineering Company. He continues “With our experience, we have analyzed that BI failure is generally not due to technology but often due lack of user support or because of poor dashboard design. For successful BI implementation, it is critical to select right users and educate them to use BI self-service features and train them to effectively analyse and interpret information”.

6. CONCLUSION

Inspired by the growing demand for dashboards for decision making, the primary goal of our paper was to examine how BI dashboards add value in decision making and performance management in the organizations and what are the challenges in implementation of BI.

This study was carried out to understand how dashboards helps organizations to handle the growing complexity of data, information, and business decisions. We tried to relate and identify the BI capabilities that help business in their intended purpose and attain the expectations of the decision makers. The research revealed that dynamic and visual dashboards provide useful business insights to the decision makers and help managers to take timely and effective business decisions. BI dashboards indeed play increasingly important role in decision making, and organizations will see more opportunity for exceptional business impact. Most participants were active users of the dashboard as well as business decision makers. After obtaining some interesting insights from various professional, the researcher personally participated in making some BI dashboards for the Organizational use in various areas to validate the claims and comments of the interviewees.

Based on the overall study, the researcher suggests that:

1. Data visualization has impact on speed & quality of decision making. So dashboard design should use right visuals and right locations. Key information should be on the top of the screen, followed by related visuals and the details should be at the bottom of the screen.
2. Drill-down, maps & self-service BI capabilities and Interactive features of dashboards are mostly appreciated by users. Instead of cluttering the screen, provide fewer number of visuals and give an option to the user to click on them to drill down to further details. This will make dashboard usable by the senior managers for the overviews and at the same time can be used by other users to drill-down to further details.
3. Predictive Analytics helps the decision makers to predict future trend based on patterns. Support from BI professionals can

be very helpful for professional forecasting dashboards.

4. Most users used dashboards for KPI measurements and performance management. The KPIs used in the supply chain operations were mainly to measure ON-TIME readiness, Quality levels and cost levels based on set targets. The number of KPIs could be from 5 to 10.
5. Top management professionals prefers dashboards in scorecard form for easy relation to strategy. These included few high level KPIs related to financial goals, customer satisfaction, Employee satisfactions & internal process performance.
6. The design & layout of dashboards seem to be more of an individual choice and self-service BI provides the needed flexibility to experienced BI users to customize as per their needs. Dashboards should be designed for easy interpretation, keeping the important insights on top & details can be provided with drill-down features at next levels.
7. Managers find that BI dashboards with integrated & reliable data allows quicker response to Customers & provides improved ability to provide enterprise information at-a-glance.

Indeed, BI makes the task of transforming huge, complex data into Actionable Information easier. Effective use of BI can greatly help professionals to make improved decisions & hence BI has become one of the key pillars of information ecosystem in today's world. The findings would enable professionals to proactively address some of the challenges for a smooth BI implementation and maximize the benefits of BI dashboards.

7. LIMITATIONS AND FUTURE RESEARCH

This study is not without limitations. Being a descriptive study, there was no quantitative analysis involved and future research shall validate these based on a survey mechanism using a wider range of industries and end users of BI. The study opens up a number of new directions for further research to unearth patterns from a specific industry perspective. The results can be further expanded by interviewing more respondents globally.

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